



Does capitalisation facilitate sarcasm comprehension?

Evidence from self-paced reading

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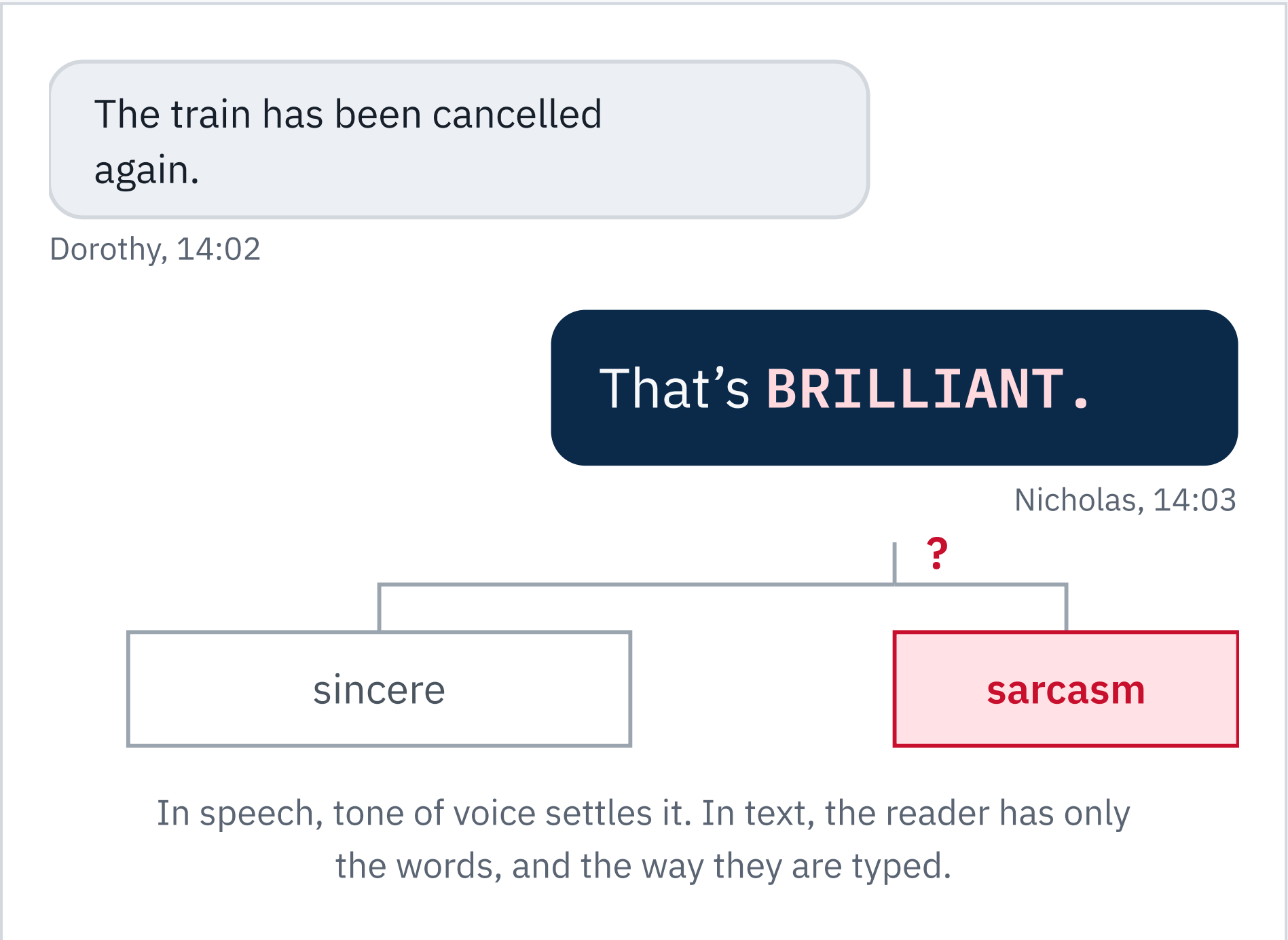
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Capitalisation slows reading, and it does *not* selectively help readers detect sarcasm.

$BF_{01} = 33\text{--}54$
Strong Bayesian evidence for the null interaction at both regions

$all\ ps > .28$
Capitalisation × Sentence Type interaction

Background



Three converging reasons to expect capitalisation to help

PRODUCTION

Writers systematically capitalise words in sarcastic and ironic contexts. (Whalen et al., 2009)

COMPUTATIONAL

Capitalisation is a significant predictive feature in automatic sarcasm detection. (Whalen et al., 2009)

THEORETICAL

Caps encode emphatic stress rather than emotion, and readers project prosodic contours onto text as they read silently. (Heath, 2018; Fodor, 2002)

1 Caps make the reader's inner prosody more emphatic.

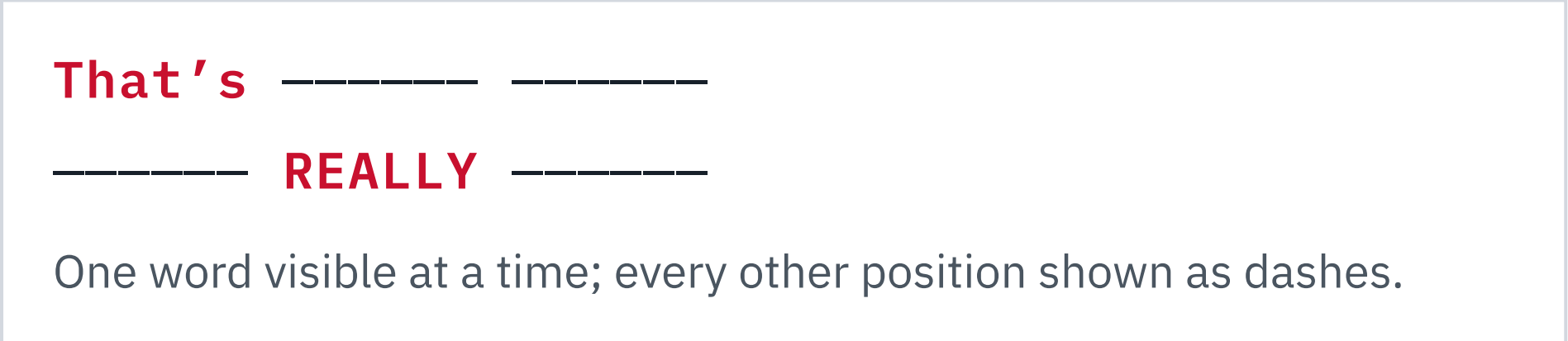
2 That emphatic contour simulates the contrastive stress of sarcastic speech, cueing non-literal meaning.

Prediction: a smaller sarcasm cost when the sarcastic phrase is capitalised.

Yet no controlled psycholinguistic experiment has tested whether capitalisation facilitates sarcasm processing during real-time reading. The present study fills that gap.

Method

Non-cumulative moving-window self-paced reading, online in PCIBex. One word per key press; reading time is the interval between presses.



113	3,512
native English speakers analysed (175 recruited)	observations at the sarcastic phrase
32	93%
items + 48 fillers, Latin-square counterbalanced	mean comprehension accuracy

Mixed-effects models (lme4) on log reading times, sum-coded factors, random structure selected with buildmer. Trials < 200 ms or > 3 SD from a participant's mean excluded (cell n = 869–894). Not pre-registered.

Design & materials

2 × 2 within-subjects; 32 items adapted from Zhu & Filik (2024) and Huang et al. (2024).



Figure 1. The two analysis regions, on the sarcastic–capitalised version of one item.

	LITERAL	SARCASTIC
NO CAPS	... really ugly and ...	... really lovely and ...
CAPITALISED	... REALLY ugly and ...	... REALLY lovely and ...

Table 1. The four conditions for the example item. The sarcastic version replaces the evaluative adjective with its semantic opposite; a comprehension question followed 20 of the 80 trials.

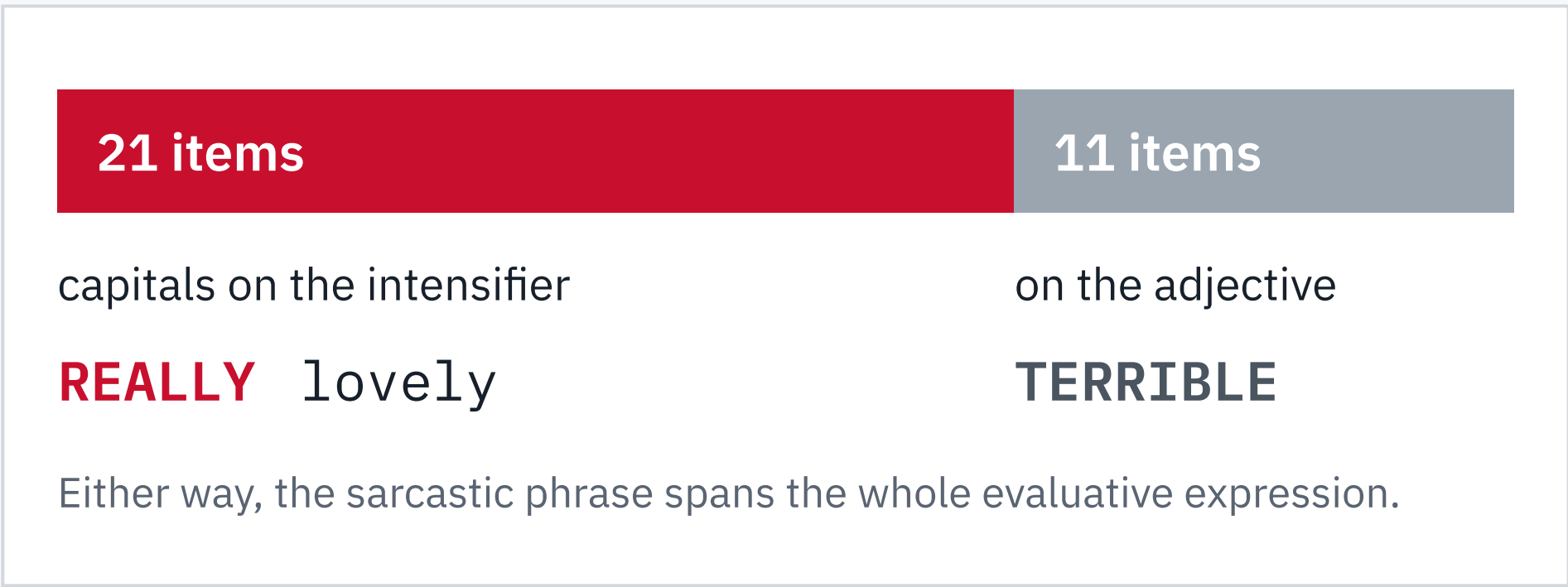


Figure 2. Where the capitals fell across the 32 items.

Capitalisation slows reading

- Readers do not ignore the typography: capitalised words cost extra time at both regions.
- The cost is the same size in sarcastic and literal sentences.

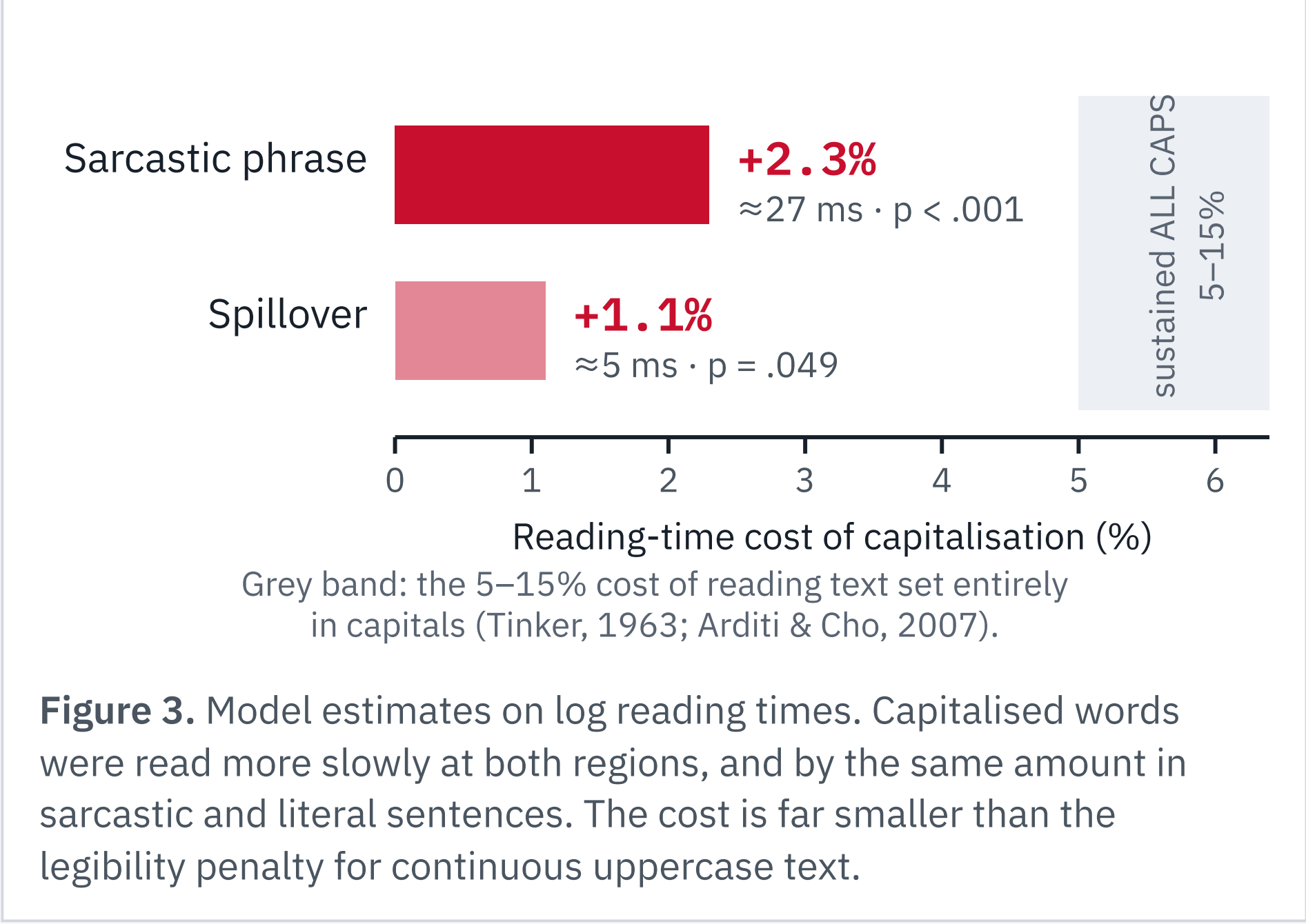


Figure 3. Model estimates on log reading times. Capitalised words were read more slowly at both regions, and by the same amount in sarcastic and literal sentences. The cost is far smaller than the legibility penalty for continuous uppercase text.

Sarcasm costs appear downstream

- At the sarcastic phrase, sarcastic and literal sentences did not differ reliably ($p = .255$). One word later the cost is robust: $\approx 3.1\%$, about 40 ms, $p = .001$.

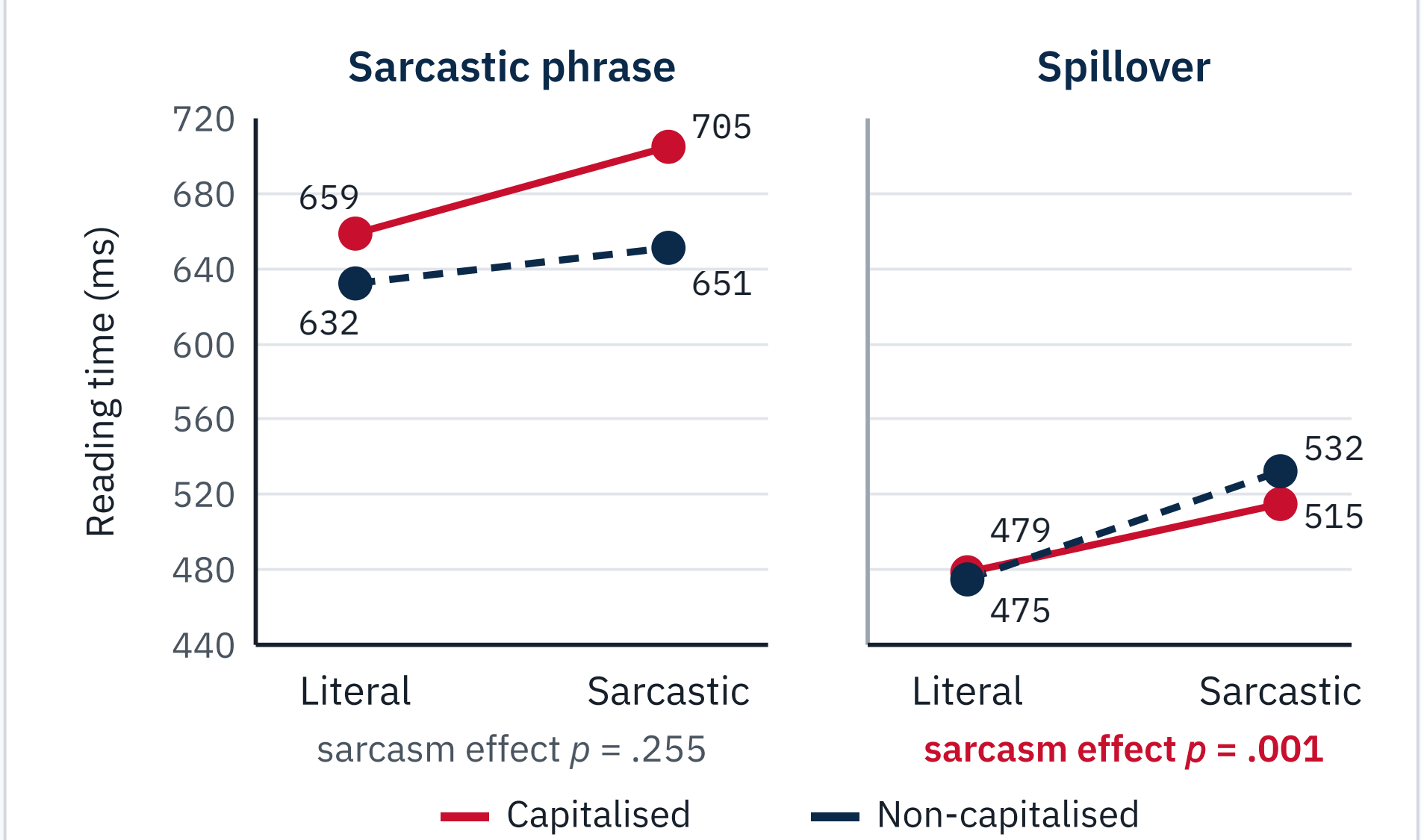


Figure 4. Raw cell means, ms, both panels on one shared scale. At the spillover the sarcasm cost looks larger without capitals, but the interaction is still null: that cell is highly skewed ($SD = 492$ ms vs $283\text{--}326$ ms elsewhere) and models are fitted to log reading times.

Evidence for the null

- The predicted interaction was absent at both regions ($p = .661$, $p = .280$), with strong Bayesian support for the null. Interactions of about 8 ms were detectable at 80% power: an absence of effect, not a failure to look.

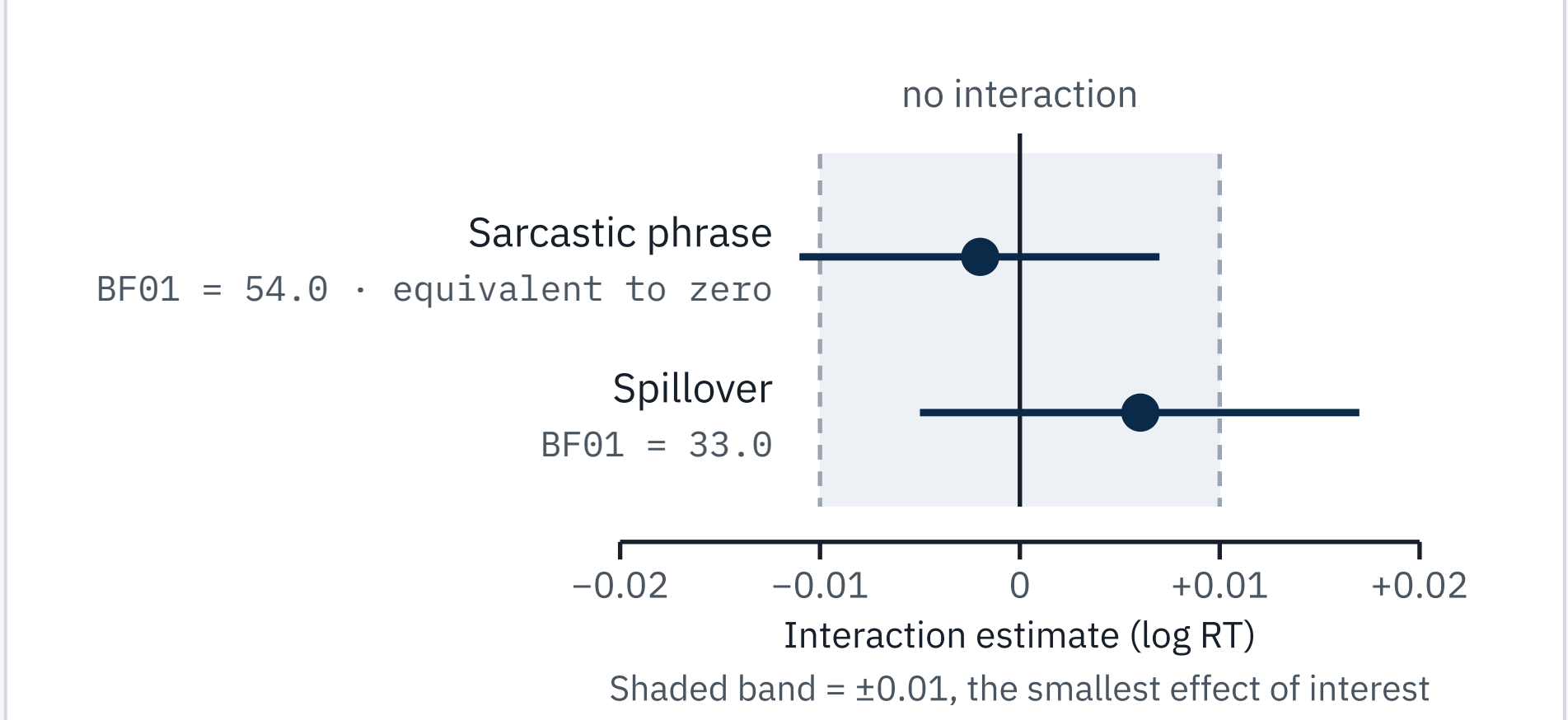


Figure 5. Capitalisation × Sentence Type estimates with 95% CIs. Equivalence to zero holds at the sarcastic phrase (TOST $p = .035$) but not at the spillover ($p = .236$); TOST uses the 90% interval, so the 95% CI can reach past the band.

Discussion

- Despite strong theoretical and production-side reasons to expect capitalisation would aid sarcasm detection, these results indicate it does not.
- It slows reading regardless of whether the sentence is sarcastic or literal, consistent with its role as a marker of **visual salience**.

Emphasis is ambiguous: one can emphatically agree (“I am SO happy”) or emphatically mock (“SO helpful”).

- Sarcasm costs emerge downstream of the critical word, in line with accounts in which pragmatic inference unfolds incrementally as the expression is integrated with context.
- The null interaction, on frequentist and Bayesian evidence alike, challenges the assumption that typographic emphasis can compensate for absent vocal prosody: readers rely on context and semantics instead.
- Caveat.** Self-paced reading removes parafoveal preview; eye-tracking would let caps be seen before fixation.

Capitalisation tells readers “pay attention here”. It does not tell them why.

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